Machine Learning and Artificial Intelligence in Computational Arts, Music, and Games

These slides are live at:
https://ktatar.github.io/2026-06-university-of-oslo

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Acknowledgements

Country	Funding Body	Timeline
Sweden	The Wallenberg AI, Autonomous Systems, and Software Program – Humanities and Society	2025-2030
Sweden	VR - Vetenskapsrådet	2025-2031
Sweden	The Wallenberg AI, Autonomous Systems, and Software Program – Humanities and Society	2021-2026
Canada	Canada Council for the Arts	2018-2021
Canada	BC Arts Council	2020
Switzerland	Swiss National Science Foundation	2020-2021
Canada	Social Sciences and Humanities Research Council	2014-2020
Canada	Natural Sciences and Engineering Research Council	2014-2019

Research Themes

- Deep Learning and Audio
- AI in Computational Creativity and Game Design
- Societal Impact of AI in Culture, Arts, and Music
- Artworks

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- **Deep Learning and Audio**
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Deep Learning and Audio

-Latent Timbre Synthesis

-Coding the Latent Artwork and RawAudio Variational Autoencoder

-Music Notation and Composition with Latent Spaces

-ARIA: A Diagnostic Framework for Music Training Data Attribution

-Neuralacoustics

-Neural Audio Instruments

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Latent Timbre Synthesis

K. Tatar, D. Bisig, and P. Pasquier, “Latent Timbre Synthesis,” Neural Computing & Applications, Oct. 2020, doi: 10.1007/s00521-020-05424-2.

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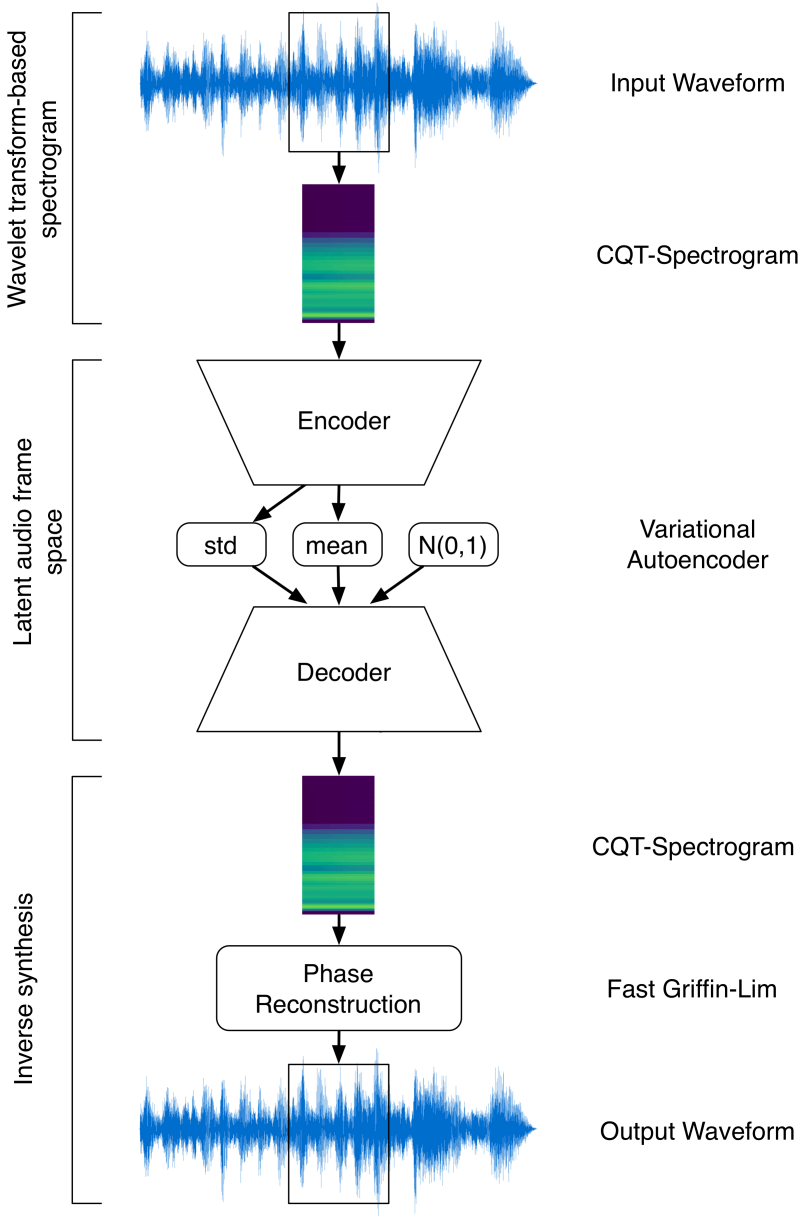
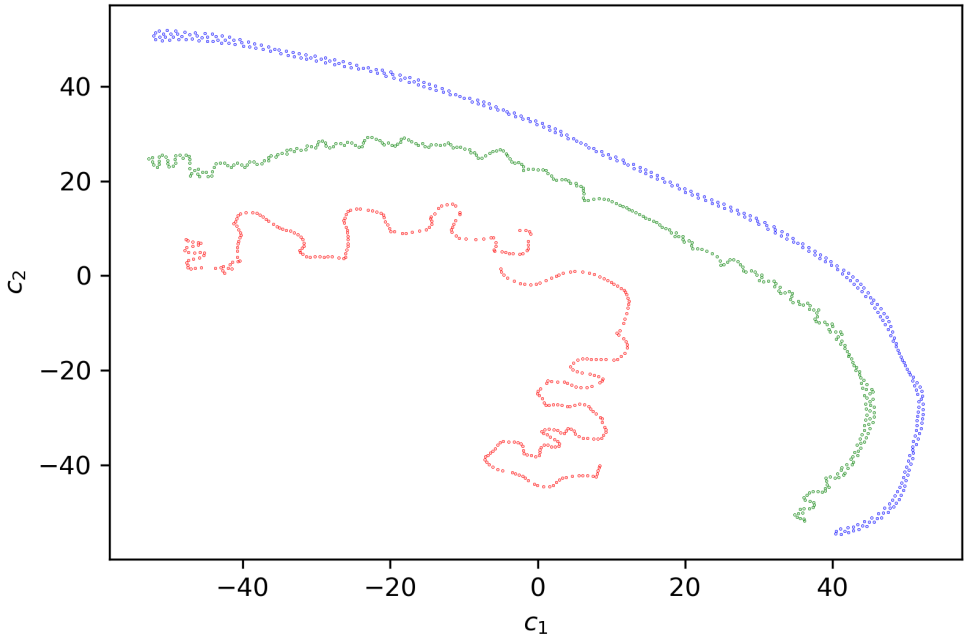
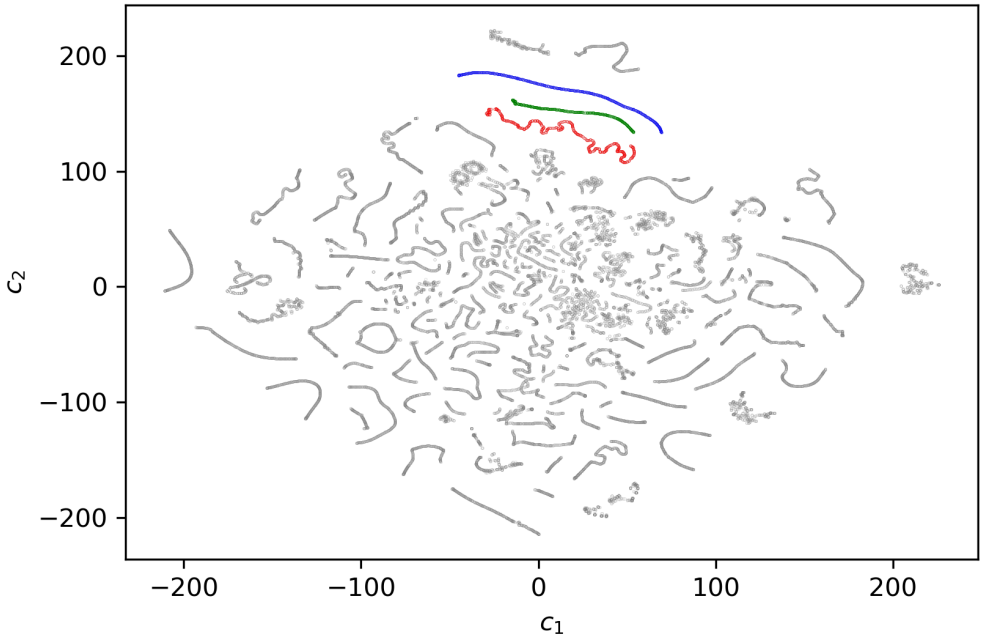


Latent Timbre Synthesis

Interpolations in the latent space of the VAE





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Latent Timbre Synthesis

https://www.youtube.com/watch?v=-1XuXbX_VZo

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gyroscope; picture-in-picture; web-share" referrerpolicy="strict-
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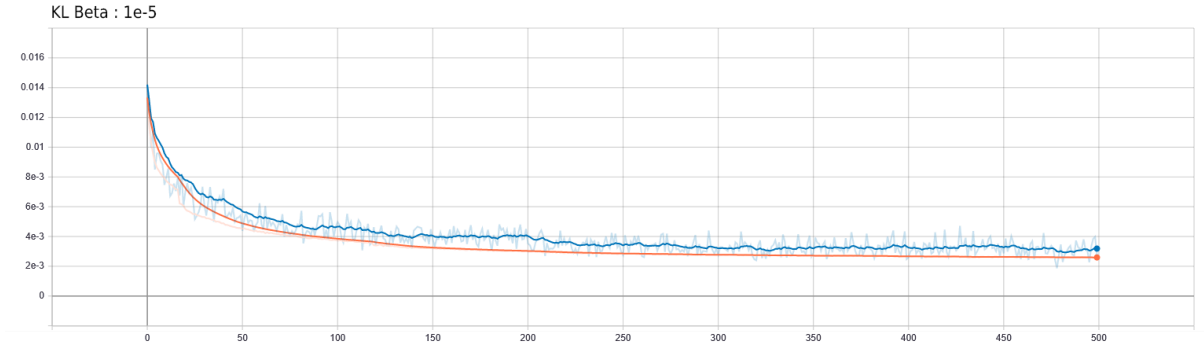
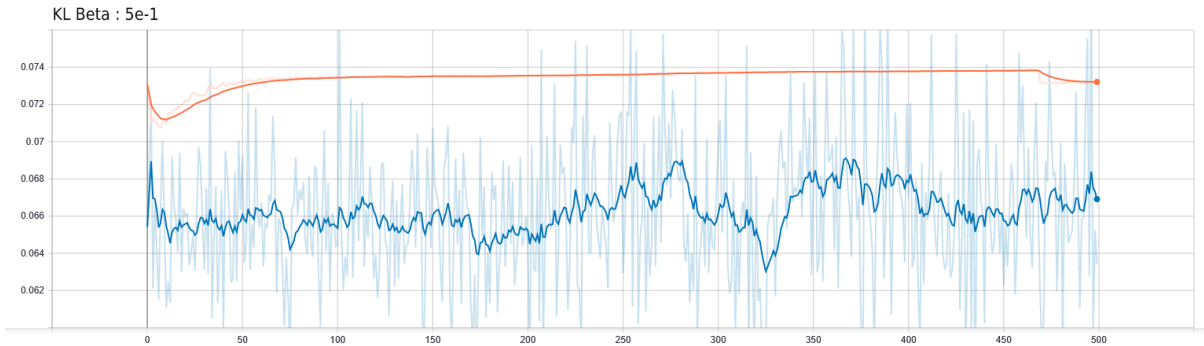
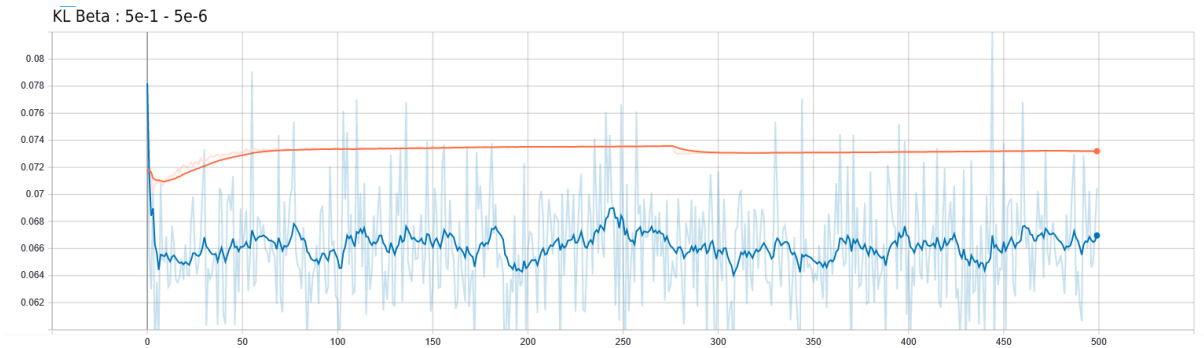
Deep Generative Modelling is not neutral.

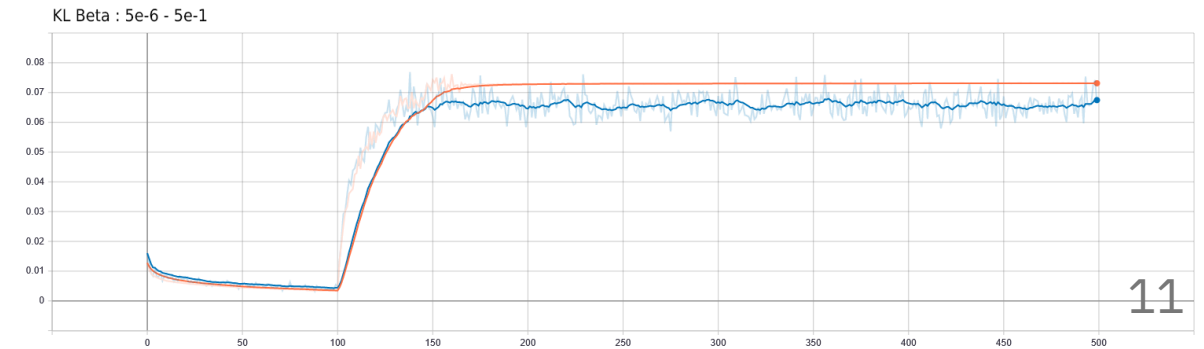
Latent Timbre Synthesis

Notes on Reproducibility









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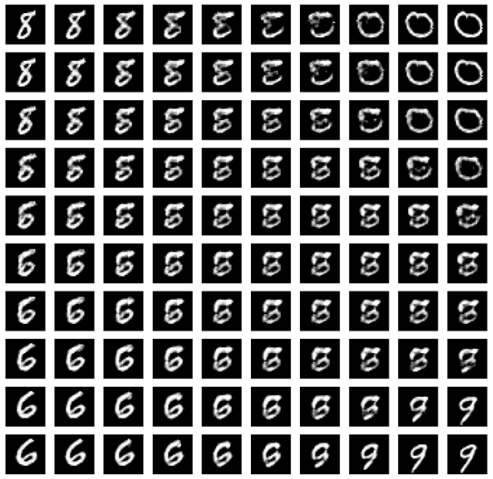
Latent Timbre Synthesis

Notes on Reproducibility

KL Beta : 1e-5



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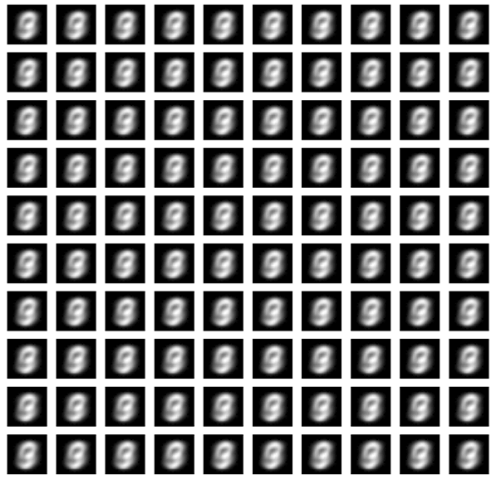
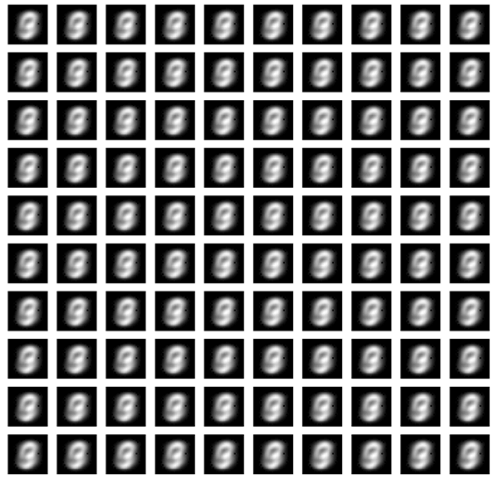
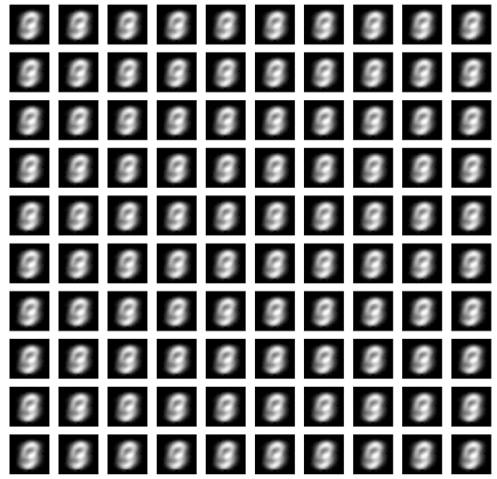


KL Beta : 5e-1 - 5e-6



KL Beta : 5e-6 - 5e-1



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Most Musical AI research are ad-hoc efforts.

Latent Timbre Synthesis

Emerging themes from the user study with 9 composers:

- -Iterative sound design in musical composition
- -Musical strategies
- -Musical goals and concepts
- -Familiarity
- -Affordances
- -Sound aesthetics
- -Sound quality
- -User Inteface
- -Tool Deficiencies
- -Continued use

Deep Learning and Audio

- Latent Timbre Synthesis
- **Coding the Latent Artwork and RawAudio Variational Autoencoder**
- Music Notation and Composition with Latent Spaces
- ARIA: A Diagnostic Framework for Music Training Data Attribution
- Neuralacoustics
- Neural Audio Instruments

Coding the Latent Artwork and RawAudio Variational Autoencoder

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si=b3xsvPWKXwDRp12e&start=2810" title="YouTube video player"
frameborder="0" allow="accelerometer; autoplay; clipboard-write;
encrypted-media; gyroscope; picture-in-picture; web-share"
referrerpolicy="strict-origin-when-cross-origin" allowfullscreen>
</iframe>
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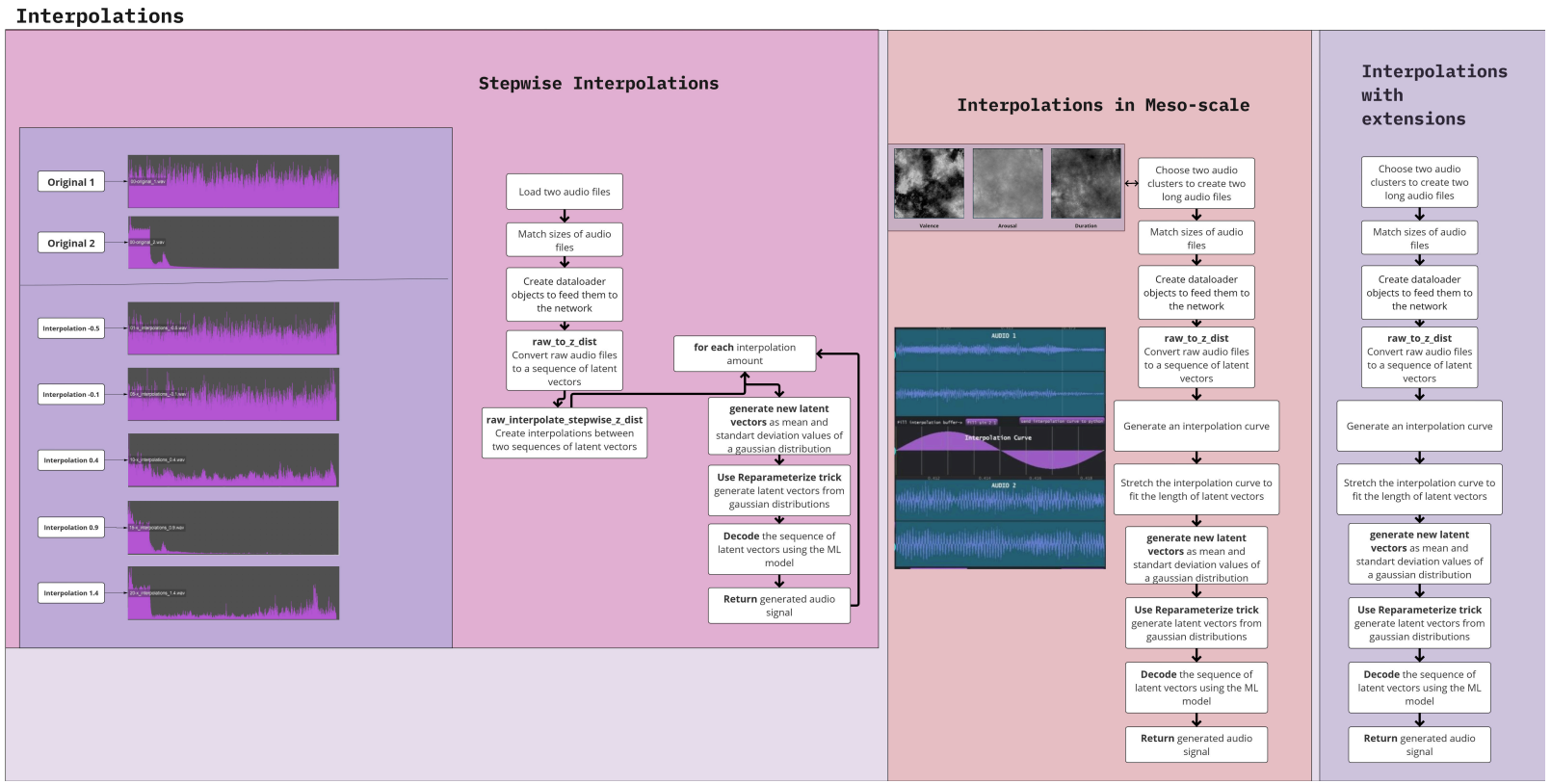
How do we make music with vast sound search spaces?

RawAudio Variational Autoencoder



K. Tatar, K. Cotton, and D. Bisig, “Sound Design Strategies for Latent Audio Space Explorations Using Deep Learning Architectures,” presented at the Proceedings of Sound and Music Computing 2023, 2023.

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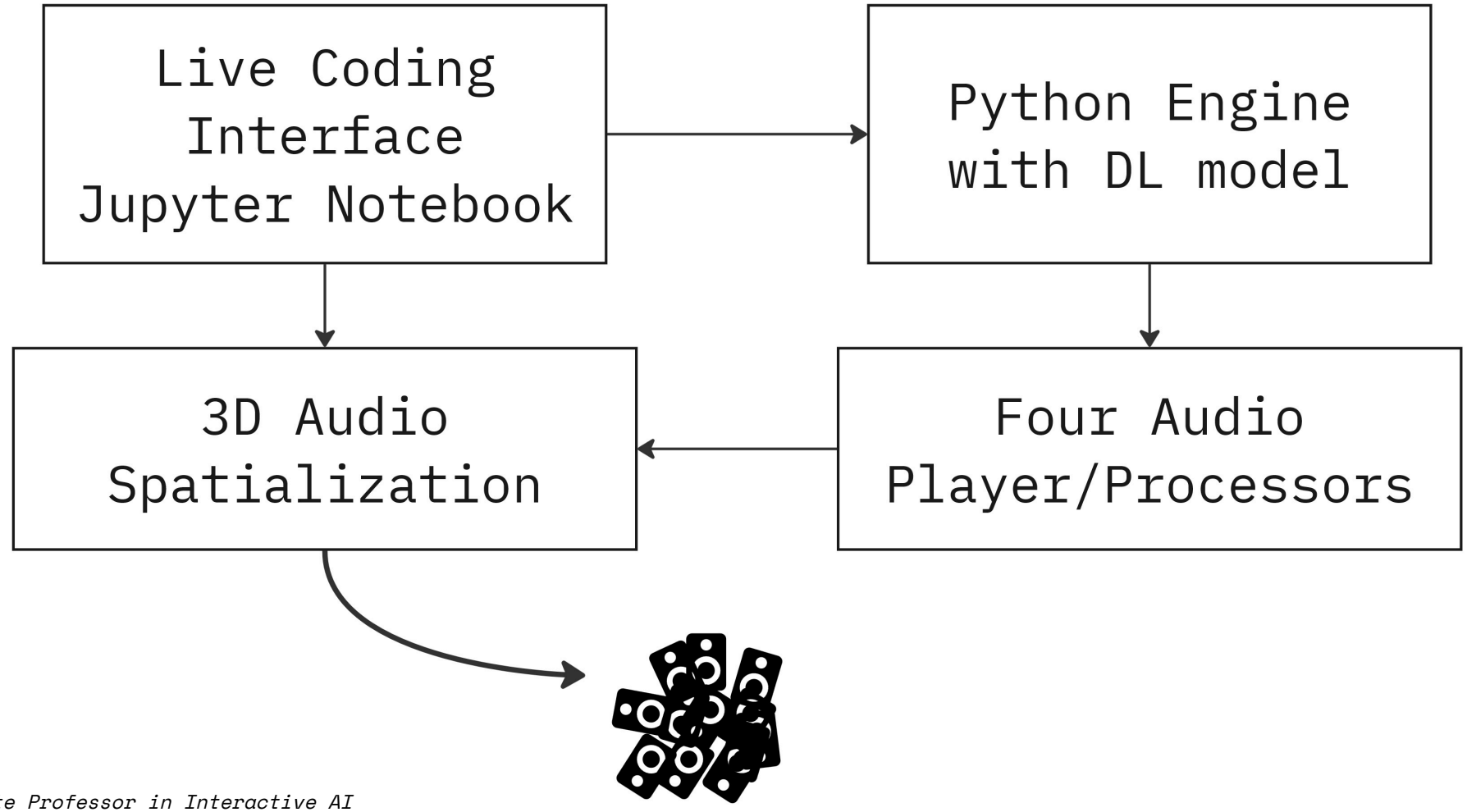
RawAudio Variational Autoencoder



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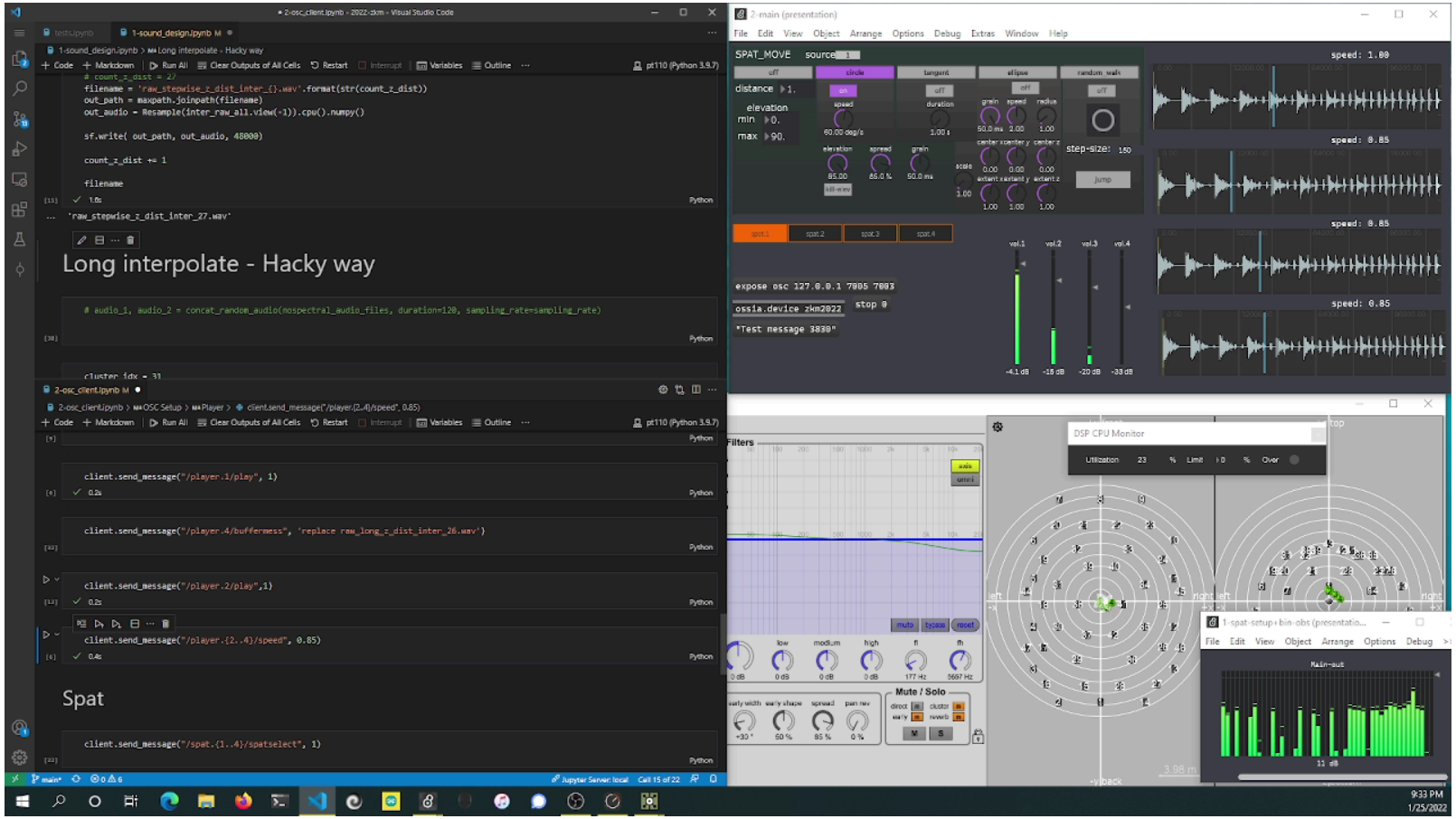
Coding the Latent



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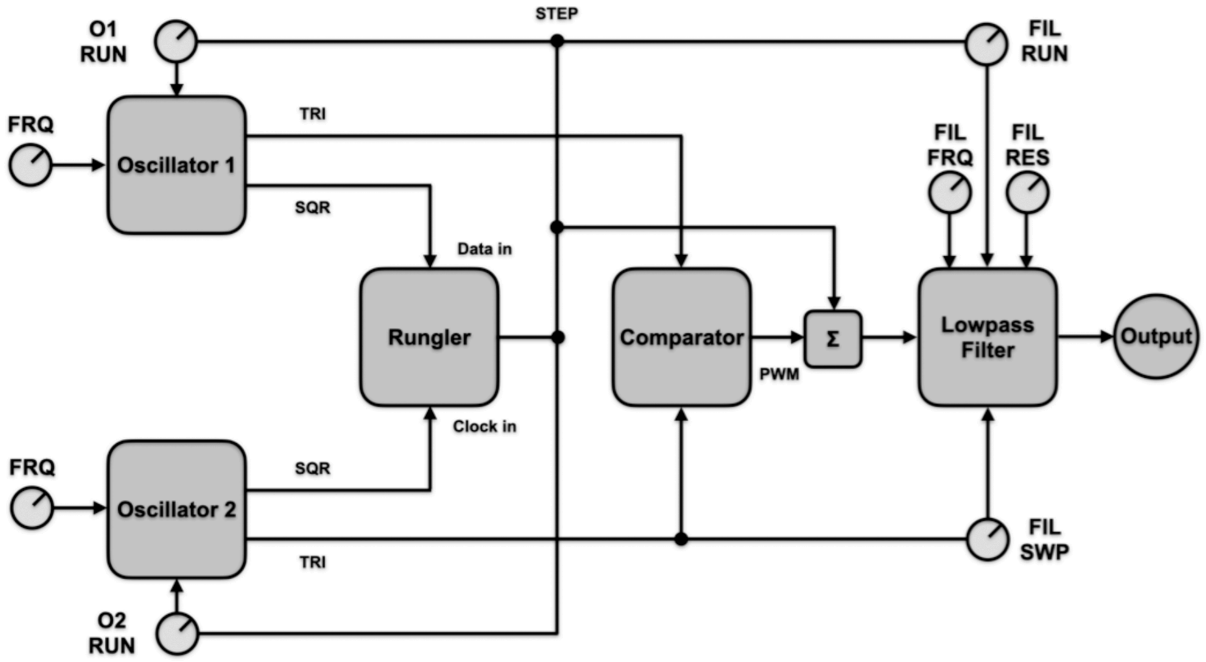
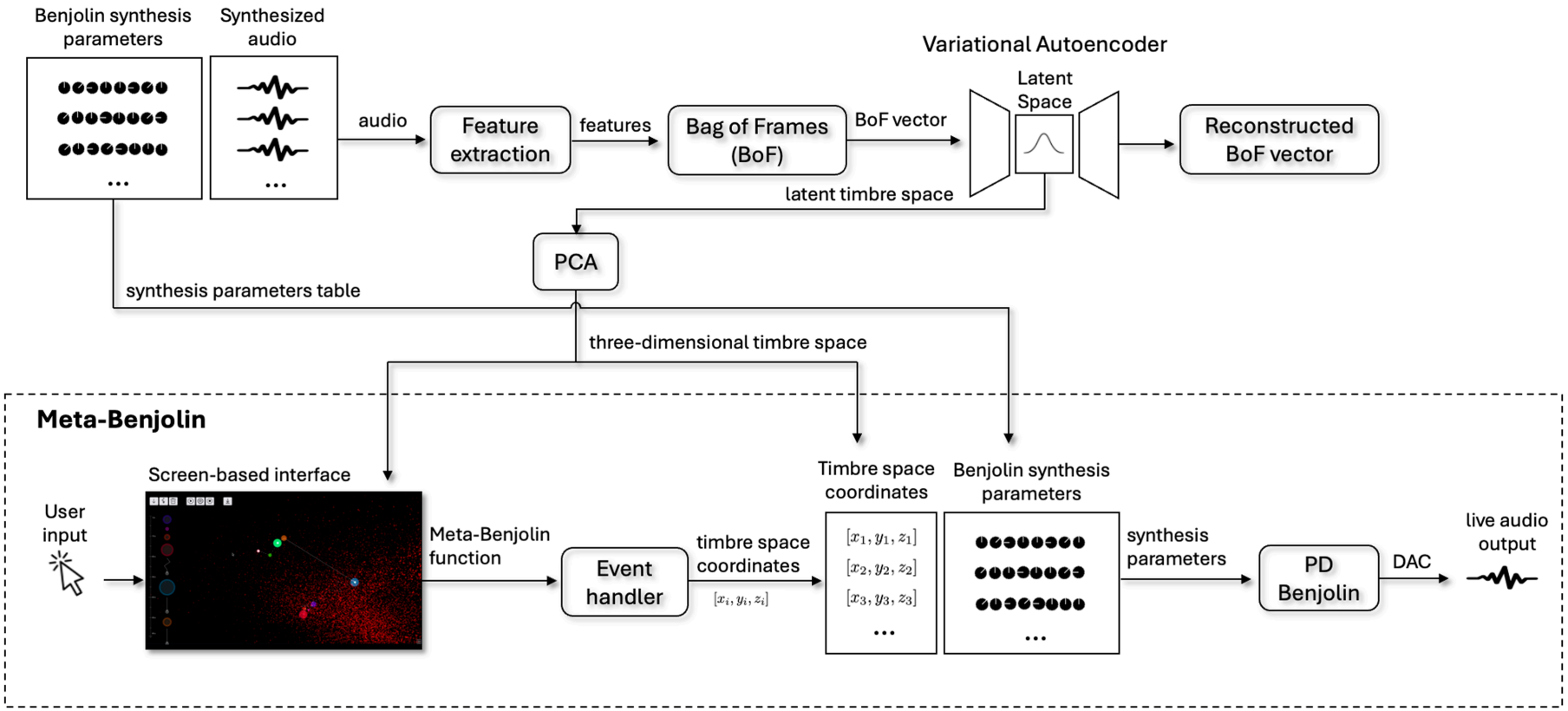
Music Notation and Composition with Latent Spaces

Meta-Benjolin



Madaghiele V., Lund L., Holzer D., Kelkar T., Tatar, K., and Holzapfel A. (2026). Expanding the machine: notating generative synthesis with a state-based representation and an interactive timbre space. Organised Sound, Cambridge Press.

Music Notation and Composition with Latent Spaces



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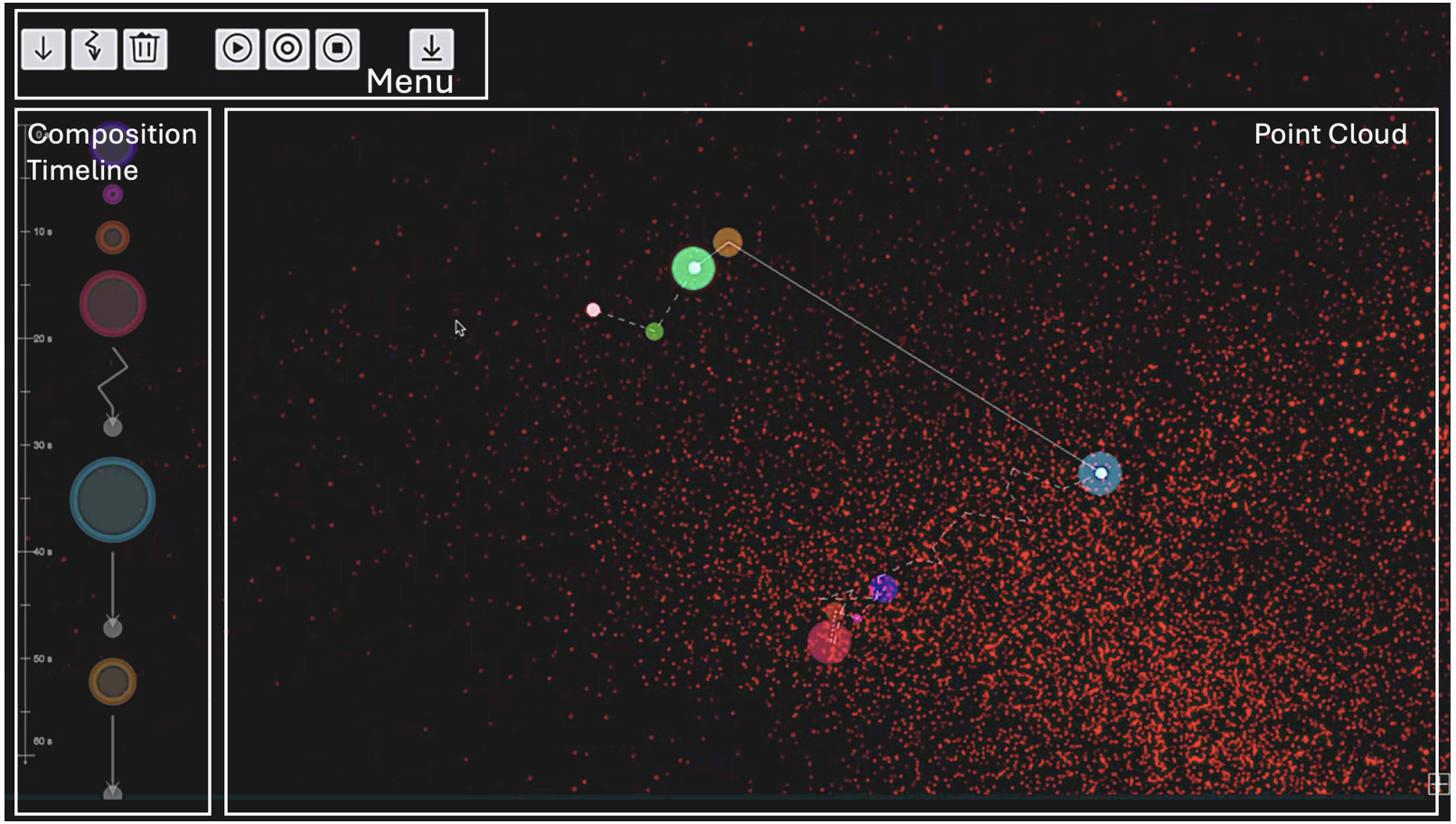
Music Notation and Composition with Latent Spaces



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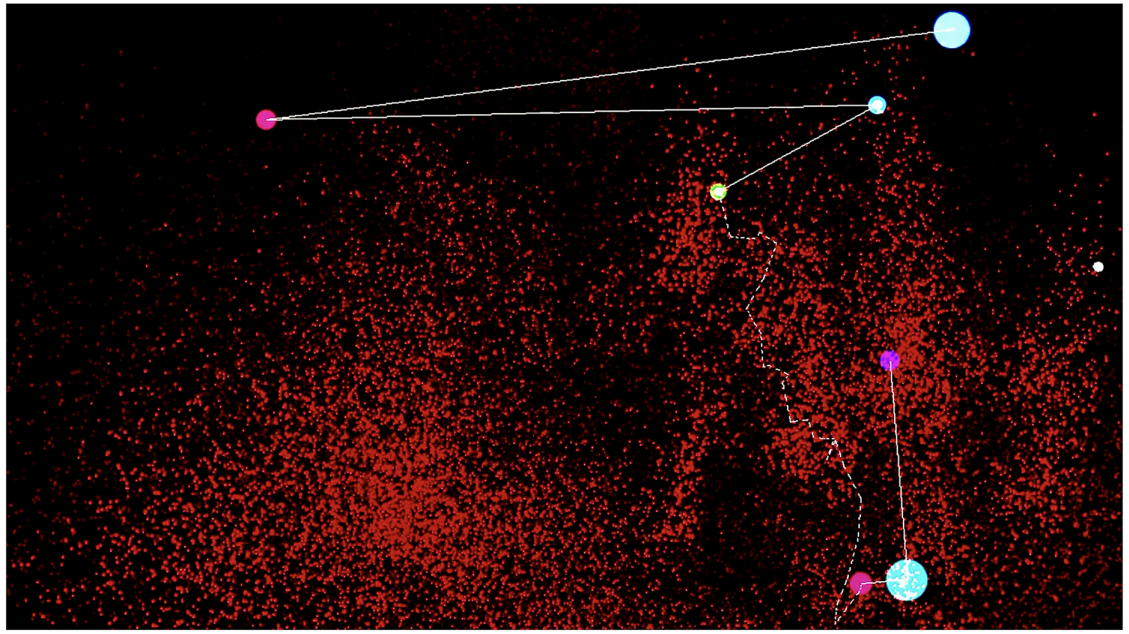
Music Notation and Composition with Latent Spaces

(a)



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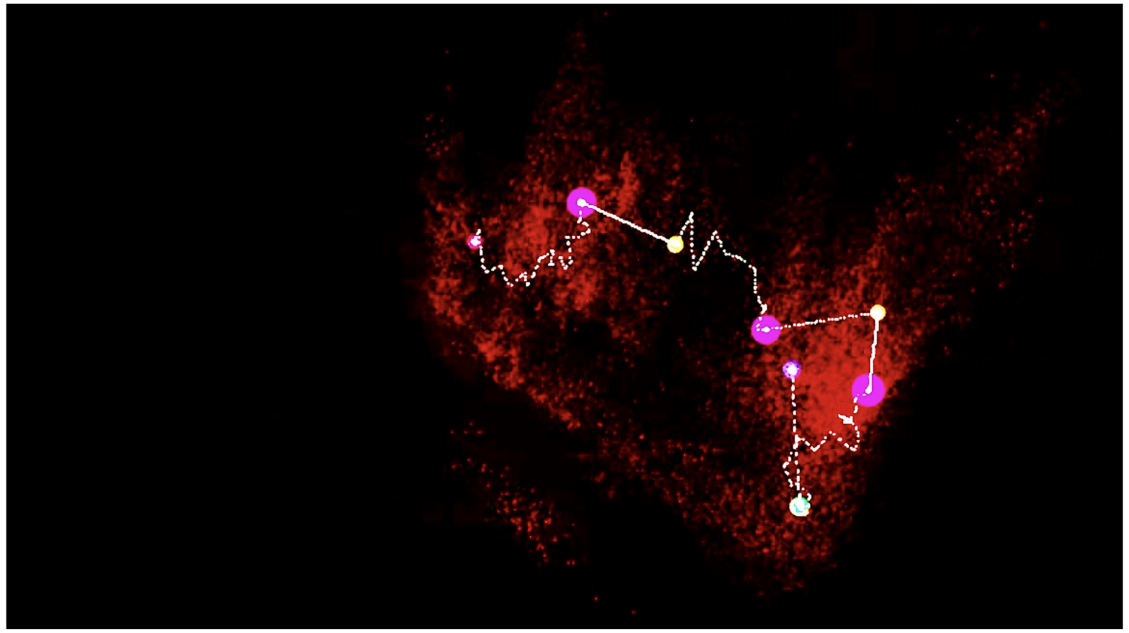

Examples of the use of transitions to navigate long distances. D2 used a meander transition in the middle of the piece to connect two sections; within a section, neighbouring states are connected using crossfades. A5 used a crossfade and a meander transition to navigate between two neighbourhoods in the cloud, each corresponding to a section in their piece. (a) Composition by D2 (detail), Sound_example_4.m4a in the sound material. (b) Composition by A5 (detail), Sound_example_5.m4a in the sound material.

Music Notation and Composition with Latent Spaces

https://meta-benjolin.com/

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downgrade" allowfullscreen ></iframe>
```

Madaghiele V., Lund L., Holzer D., Kelkar T., Tatar, K., and Holzapfel A. (2026). Expanding the machine: notating generative synthesis with a state-based representation and an interactive timbre space. Organised Sound, Cambridge Press.

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-Neural Audio Instruments

ARIA: A Diagnostic Framework for Music Training Data Attribution

Current training data attribution (TDA) provides a single scalar value to quantify the influence of each training example on a model's prediction.

Music generation and its copyright analysis require two additional specifications:

1- **Which training data** (audio, music, composition etc.) influence a generated output
2- Along **which musical aspects** the influence operates

Han, C., Panahi, A., & Tatar, K. (2026). ARIA: A Diagnostic Framework for Music Training Data Attribution. Submitted to NeurIPS 2026. https://arxiv.org/pdf/2605.16181

Recap: Datamodels

How much does each individual training data influence the prediction (or model output) with a new input?

Definition 1 (Datamodeling). *Consider a fixed training set S, a learning algorithm $\mathcal{A}$, a target example x, and a distribution $\mathcal{D}_S$ over subsets of S. For any set $S' \subset S$, let $f_{\mathcal{A}}(x; S')$ be the (stochastic) output of training a model on S' using $\mathcal{A}$, and evaluating on x. A datamodel for x is a parametric function g_θ optimized to predict $f_{\mathcal{A}}(x; S_i)$ from training subsets $S_i \sim \mathcal{D}_S$, i.e.,*

$$g_\theta : \{0,1\}^{|S|} \to \mathbb{R}, \qquad \text{where} \qquad \theta = \arg\min_w \widehat{\mathbb{E}}^{(m)}_{S_i \sim \mathcal{D}_S} \left[\mathcal{L}\left(g_w(\mathbf{1}_{S_i}),\, f_{\mathcal{A}}(x; S_i) \right) \right],$$

$\mathbf{1}_{S_i} \in \{0,1\}^{|S|}$ is the characteristic vector of S_i in S (see (3)), $\mathcal{L}(\cdot,\cdot)$ is a loss function, and $\widehat{\mathbb{E}}^{(m)}$ is an m-sample empirical estimate of the expectation.

Ilyas, A., Park, S. M., Engstrom, L., Leclerc, G., & Madry, A. (2022). Datamodels: Predicting predictions from training data. Proceedings of the 39 th International Conference on Machine Learning, Baltimore, Maryland, USA, PMLR 162, 2022.

Algorithm A.1 An outline of the datamodeling framework: we use a simple parametric model as a proxy for the entire end-to-end training process.

procedure ESTIMATEDATAMODEL(target example x, trainset S of size d, subsampling frac. $\alpha \in (0,1)$)
  $T \leftarrow []$ ▷ Initialize *datamodel training set*
  for $i \in \{1, \ldots, m\}$ **do**
    Sample a subset $S_i \subset S$ from $\mathcal{D}_S$ where $|S_i| = \alpha \cdot d$
    $y_i \leftarrow f_{\mathcal{A}}(x; S_i)$ ▷ Train a model on S_i using $\mathcal{A}$, evaluate on x
    Define $\mathbf{1}_{S_i} \in \{0,1\}^d$ as $(\mathbf{1}_{S_i})_j = 1$ if $x_j \in S_i$ else 0
    $T \leftarrow T + [(\mathbf{1}_{S_i}, y_i)]$ ▷ Update datamodel training set
  $\theta \leftarrow$ RUNREGRESSION(T) ▷ Predict the y_i from the $\mathbf{1}_{S_i}$ vectors
  return θ ▷ Result: a weight vector $\theta \in \mathbb{R}^d$

Algorithm A.2 Assessing datamodels' ability to predict dataset counterfactuals.

procedure COUNTERFACTUALEVAL(target example x, datamodel θ, trainset of size d)
  $M_0 \leftarrow$ AVERAGE$[f_{\mathcal{A}}(x; [d])$ **for** $i \leftarrow 1 \ldots 100]$ ▷ Average margin on x with full training set
  for $k \in \{20, 40, 80, 160, 320\}$ **do**
    $G \leftarrow$ TOP-K(θ, k) ▷ Get the top-k indices of θ
    $M \leftarrow$ AVERAGE$[f_{\mathcal{A}}(x; [d] \setminus G)$ **for** $i \leftarrow 1 \ldots 20]$ ▷ Average margin without G over 20 trials
    $\Delta_{avg} \leftarrow M_0 - M$ ▷ Actual counterfactual
    $\widehat{\Delta} \leftarrow \theta^\top \mathbf{1}_G \; (= \sum_{i \in G} \theta_i)$ ▷ Datamodel-predicted counterfactual
  Compare all values of $\widehat{\Delta}$ and Δ_{avg}

Ilyas, A., Park, S. M., Engstrom, L., Leclerc, G., & Madry, A. (2022). Datamodels: Predicting predictions from training data. Proceedings of the 39 th International Conference on Machine Learning, Baltimore, Maryland, USA, PMLR 162, 2022.

ARIA: A Diagnostic Framework for Music Training Data Attribution

Two structural limits of data attribution in music generation:

1- **Infeasibility of counterfactual retraining**: Conventional data attribution with retraining at the scale of musical audio generation models is computationally infeasible.
2- **Music similarity is multidimensional**: Single scalar cannot support such evidence under the courts' idea-expression distinction.

These limits matter because, with music copyright lawsuits actively underway, attribution evidence linking model outputs to specific training data has become increasingly important.

Han, C., Panahi, A., & Tatar, K. (2026). ARIA: A Diagnostic Framework for Music Training Data Attribution. Submitted to NeurIPS 2026. https://arxiv.org/pdf/2605.16181

Case 1: Text-to-Audio Generation with Open-MusicLM

MusicLM-style three-stage hierarchical musical audio generation model trained on FMA-Large. The hierarchical structure provides three independently attributable stages: **semantic, coarse, and fine**; each predicting a different token type from a different audio segment duration, which allows us to measure whether each stage's attribution recovers a different musical aspect.

Case 2: Symbolic Music Generation with MusicTransformer

A pre-trained MusicTransformer on MAESTRO with an Linear Datamodeling Score (LDS) attribution as the ground truth, run four attribution evaluation methods: TRAK (10-ensemble), TracIn , GRAD-COS, and GradDot; through the ARIA's evaluation pipeline, obtaining score matrixes. We use the five jSymbolic 2.2 features (**melody, harmony, rhythm, dynamic, texture**), extracted per training segment from the decoded MIDI.

Case 1: Text-to-Audio Generation with Open-MusicLM

Applying ARIA to a MusicLM-style musical audio generation model, embedding retrieval baselines diverge along the lines of each encoder's pretraining objective, indicating that they report what the encoder represents rather than the influence of training data on the generative model.

Case 2: Symbolic Music Generation with MusicTransformer

All four reliability diagnostics rank attribution methods identically to LDS, supporting their use as a substitute signal where LDS is intractable.

Han, C., Panahi, A., & Tatar, K. (2026). ARIA: A Diagnostic Framework for Music Training Data Attribution. Submitted to NeurIPS 2026. https://arxiv.org/pdf/2605.16181

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Neuralacoustics

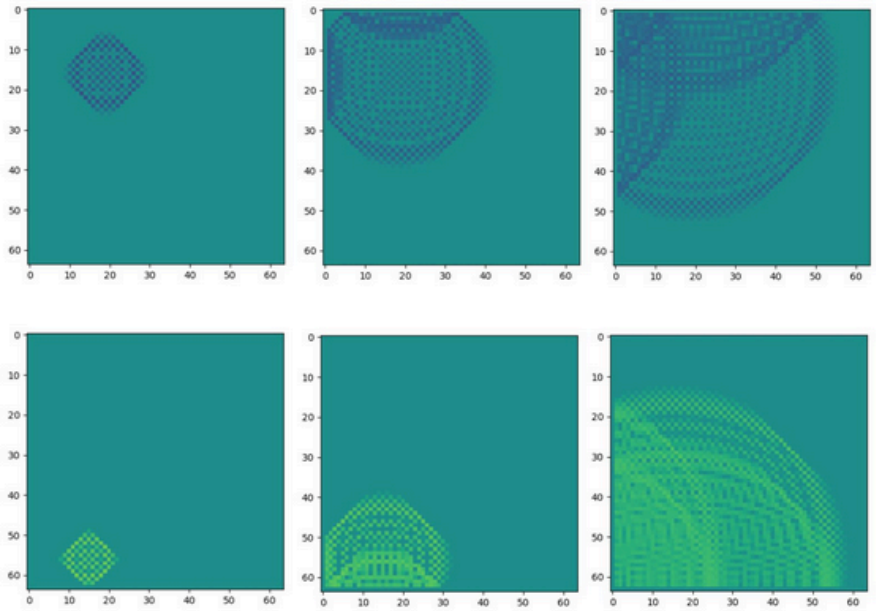

Figure 1

Visualization of a 2-dimensional membrane model. Under different excitations, the top and the bottom row each shows three snapshots of the evolution of the membrane's displacements over time.

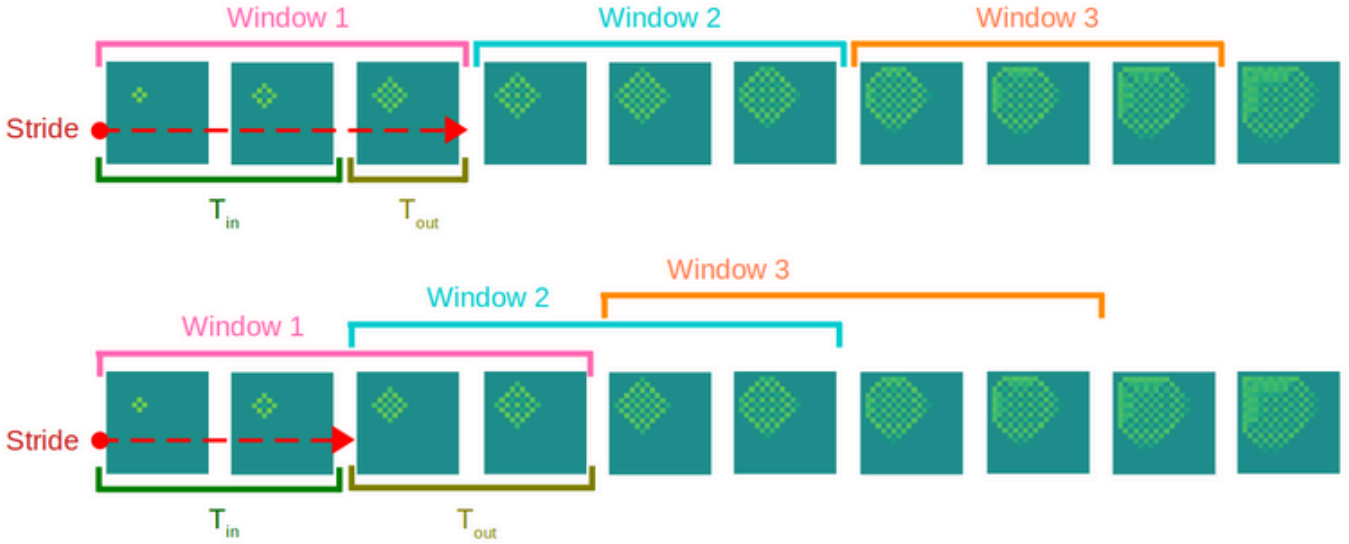

Figure 4

Windowing algorithm used to extract data points from Neuralacoustics datasets. The same dataset entry is processed via two different sets of parameters, to exemplify the extraction of different data points. The diagram depicts only the solution time steps (acoustic fields) stored in the entry's dictionary.

Chen, J., Tatar, K., & Zappi, V.. (2024). A Deep Learning Framework for Musical Acoustics Simulations. In Proceedings of the AI Music Creativity Conference. Oxford, London, September 2024. https://aimc2024.pubpub.org/pub/5cl1cvmy/release/1

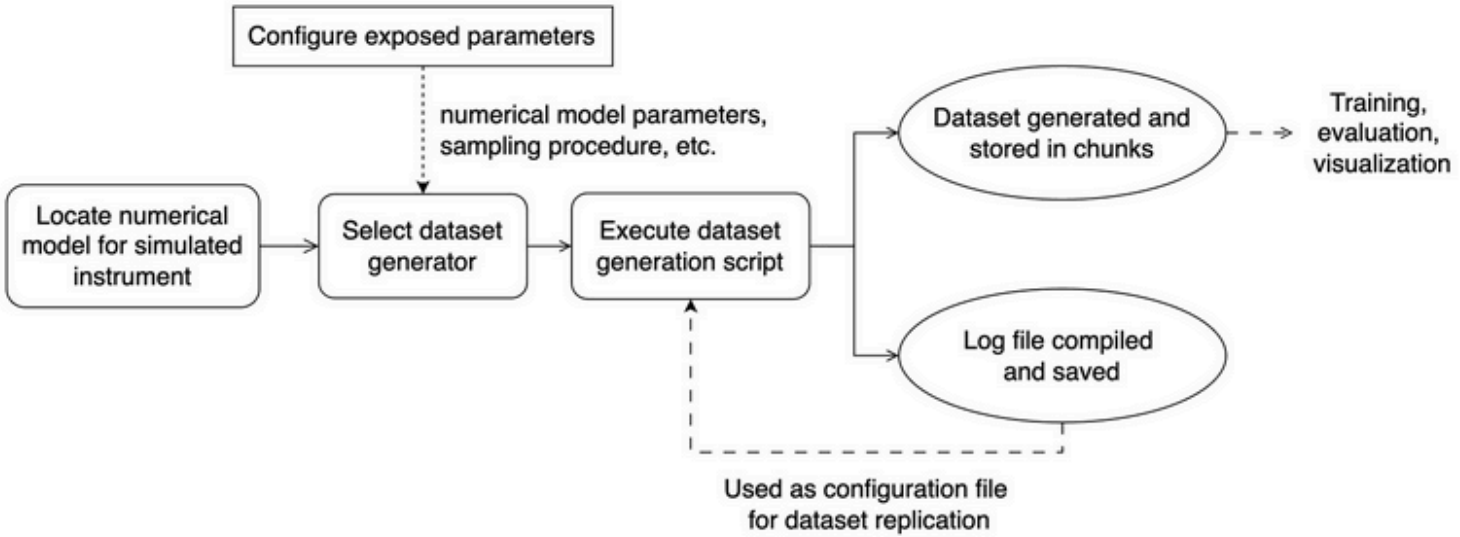

Figure 3
Flowchart for dataset generation process.

Neuralacoustics

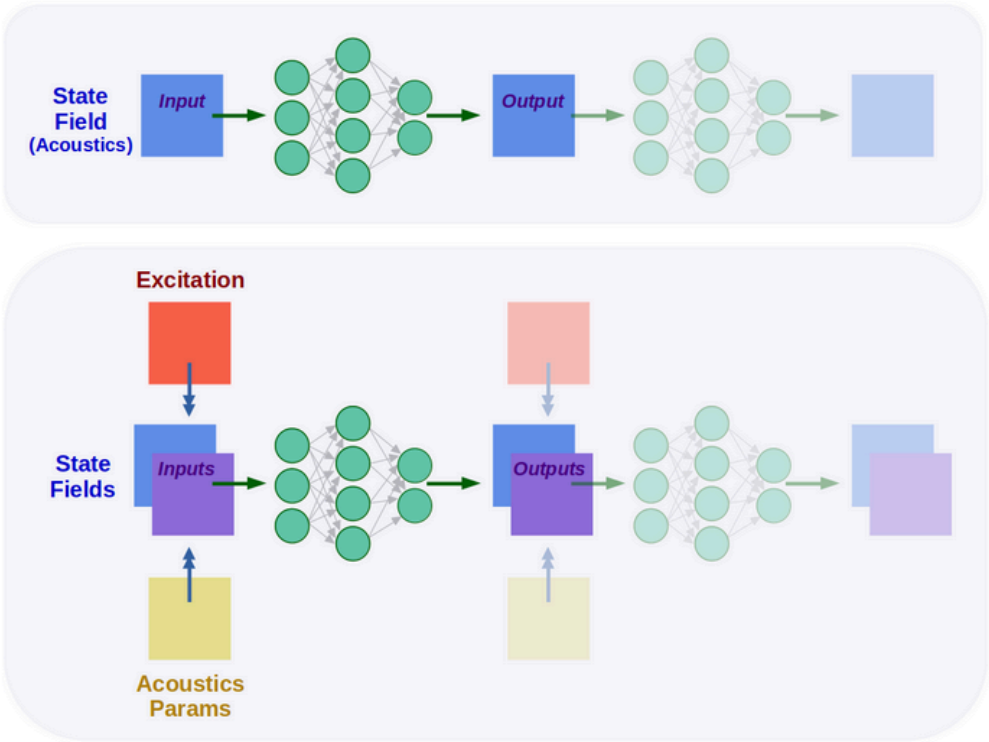

Figure 2
Neural network approximating the mapping between input/output states representing the same physical quantity (top); neural network approximating the mapping between heterogeneous input/output states, driven by continuous excitation and dynamic acoustics parameters (bottom). Only the first working scenario has been explored in the literature.

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-**Neural Audio Instruments**

Neural Audio Instruments

Neural Audio is a category of deep learning pipelines which output audio signals directly, in real-time scenarios of action-sound interactions

We define neural audio instruments as Digital Musical Instruments that embed neural network and deep learning approaches capable of directly generating or transforming audio signals, and enabling real-time action–sound mapping.

Zappi, V., & Tatar, K. (2025). Neural audio instruments: Epistemological and phenomenological perspectives on musical embodiment of deep learning. Frontiers in Computer Science, 7. https://doi.org/10.3389/fcomp.2025.1575168

Neural Audio Instruments

The six takeaways from our conceptualization of neural audio instruments:

- -Stand on the shoulders of giants
- -Search for new modes of interaction
- -Challenge dualities
- -Embrace inexplicability (with a grain of salt)
- -Make AI inconspicuous

Neural audio instruments: Epistemological and phenomenological perspectives on musical embodiment of deep learning. Frontiers in Computer Science, 7. https://doi.org/10.3389/fcomp.2025.1575168

0.Stand on the shoulders of giants

- -All the core insights from the DMI literature remain relevant
- -Challenges like the control bottleneck and the symbolic nature of action-to-sound can become more pronounced under AI/ML conditions
- -Established guidance on fostering embodiment in DMIs still applies here as a vital starting point!

1.Search for new modes of interaction.

- The behaviors and “materials” of any instrument strongly condition how musicians interact with it
- Neural networks, however, may exhibit properties not easily paralleled in earlier instruments.
- Novel paradigms, such as directly “traversing” multi-dimensional latent spaces, might offer fresh avenues for mapping movement and cognition to sonic outcomes

2.Challenge dualities

-A pressing and practical concern for DMIs lies in the traditional control-synthesis divide and the predicate of mapping.

-We do not suggest abandoning mapping altogether; exploration of how gesture connects to sound is a valuable design tool.

-We advocate a holistic design perspective where sound and gesture are conceived as a unified entity from the outset, rather than as two separate “containers” later bound by mapping

3.Embrace inexplicability (with a grain of salt).

-While research on explainable AI is undoubtedly worthwhile, non-explainability can play a significant role in the use and design of neural audio instruments.

-Performers and even designers of neural audio systems may choose to focus on musical outcomes rather than dissecting every underlying process.

4. Make AI inconspicuous.

- When the AI is not intended to act as a distinct musical agent, making its presence explicit may be unnecessary or even counterproductive.
- Designers might treat neural audio models as just another invisible part of the instrument's anatomy, like the string of a piano or the integrated circuit of an analog synthesizer.
- By letting the model manifest itself only through the embodiment of the musician's actions and intentions, the performer can experience a unified instrument rather than a model endowed with conspicuous (artificial) intelligence
- By rendering the model seamlessly integral, designers promote an experience of playing an instrument rather than interfacing with an AI model.

Research Themes

- Deep Learning and Audio
- **AI in Computational Creativity and Game Design**
- Societal Impact of AI in Culture, Arts, and Music
- Artworks

AI in Computational Creativity and Game Design

-Towards Computationally Creative Game Design

-LLM-Based Executable Synthesis of Goal Playable Game Design Patterns

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AI in Computational Creativity and Game Design

- **Towards Computationally Creative Game Design**
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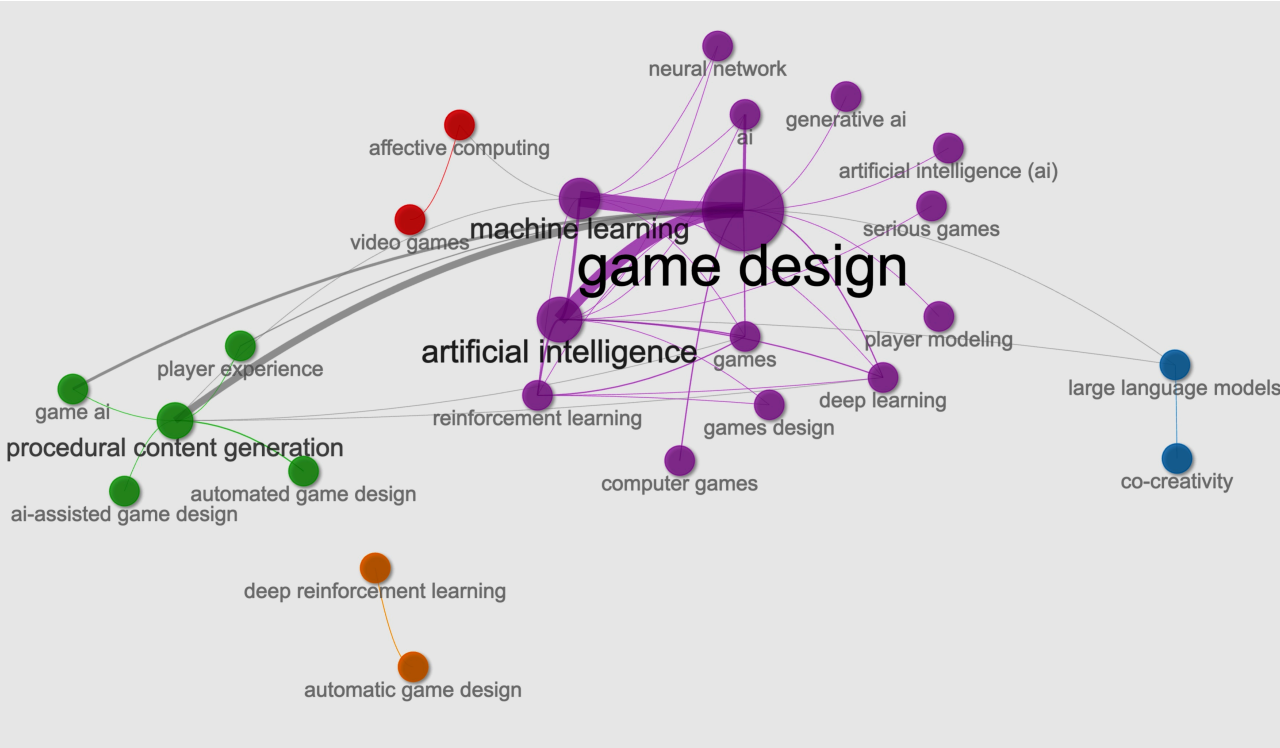
Towards Computationally Creative Game Design

Reviewed 236 studies on learning-based AI in game design

While AI technologies are widely applied for functional and procedural tasks, **creativity remains underexplored**

We investigated the correlation between the appearance of creativity-related terms and their co-occurrence with other keywords.

The co-occurrence network here visualizes the relationships and connections between frequently co-occurring keywords within the same article.



Liu, H. X., Cotton, K., Björk, S., Tatar, K. Towards Computationally Creative Game Design in Human-Computer Interaction: A Systematic Overview of Learning-based Artificial Intelligence in Game Design. Submitted to Games: Research and Practice.

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Previous work

Game Design Patterns (Bjork and Holopainen, 2005) includes 200 patterns.

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```

http://virt10.itu.chalmers.se/index.php/Category:Patterns
Bjork, S., and Holopainen, J. 2005. Patterns in game design, volume 11. Charles River Media Hingham.

AI in Computational Creativity and Game Design

- Towards Computationally Creative Game Design
- **LLM-Based Executable Synthesis of Goal Playable Game Design Patterns**

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LLM-Based Executable Synthesis of Goal Playable Game Design Patterns

We investigated whether large language models can translate structured game-design knowledge—specifically goal-pattern abstractions—into executable Unity game scenes

Game Development Framing: Digital game development as a computational creativity activity, in which design-pattern abstractions and game engine constraints guide the creation of an executable, playable game

Execution-grounded evaluation pipeline: An end-to-end workflow (LLM generation → Unity batch compilation → log-based failure analysis) for assessing executable viability at scale.

Insights on human–machine knowledge boundaries: Game design pattern knowledge injection increases structural complexity requirement at generation, revealing a tension in how domain knowledge should be injected into generative systems.

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Grounding Machine Creativity in Game Design Knowledge Representations

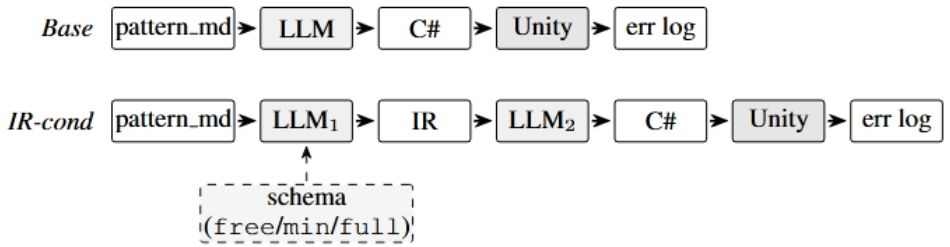

Figure 1: Baseline and IR-conditioned pipelines. Data (white), LLM (light grey), compile target (dark grey), schema injection (dashed). Both share the same Unity batch-replay harness and error-log aggregation.

Liu, H. X., & Tatar, K. (2026). Grounding Machine Creativity in Game Design Knowledge Representations: Empirical Probing of LLM-Based Executable Synthesis of Goal Playable Patterns under Structural Constraints. arXiv preprint arXiv:2603.07101.

Mage: Multi-Axis Evaluation of LLM-Generated Executable Game Scenes Beyond Compile-Pass Rate

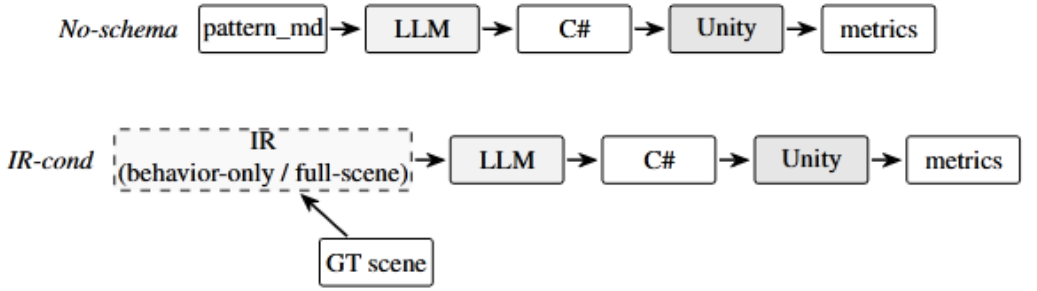

Figure 1: Two evaluation conditions. *No-schema*: the LLM receives only the natural-language pattern description. *IR-cond*: the LLM receives a ground-truth-derived IR at one of two granularity levels (behavior-only or full-scene), extracted offline from the corresponding Unity scene by an automated pipeline (Section 3.3); the natural-language description is not provided. Both conditions feed into the same Unity batch-replay harness for metric collection. Box shading: data (white), LLM (light grey), runtime artifact (dark grey), schema (dashed).

Liu, H. X., & Tatar, K. (2026). Mage: Multi-Axis Evaluation of LLM-Generated Executable Game Scenes Beyond Compile-Pass Rate. Submitted to NeurIPS 2026. arXiv preprint arXiv:2605.07342.

Mage: Multi-Axis Evaluation of LLM-Generated Executable Game Scenes Beyond Compile-Pass Rate

Table 2: Compile and runtime success rates per model and IR condition. C = compile success; R = runtime success. Counts in parentheses; runtime success is a strict subset of compile success. Qwen2.5 × full-scene is excluded due to 32K context overflow.

	No-schema		Behavior-only		Full-scene	
	C	R	C	R	C	R
Qwen3-30B	83% (65)	**80% (63)**	42% (33)	37% (29)	47% (37)	42% (33)
Qwen2.5-7B	74% (58)	39% (31)	3% (3)	3% (3)	-	-
DeepSeek-16B	23% (18)	18% (14)	29% (23)	16% (13)	42% (33)	21% (17)
Gemma-4-26B	34% (27)	34% (27)	1% (1)	1% (1)	0% (0)	0% (0)
Mean	54%	43%	19%	14%	29%	21%

Table 5: Top 10 compile-error codes across the 858 evaluation attempts. NS = no-schema, BO = behavior-only, FS = full-scene.

Code	Meaning	Total	NS	BO	FS
CS0122	Inaccessible due to protection level	991	12	862	117
CS0029	Cannot implicitly convert type	730	219	291	220
CS0136	Local variable name conflict (scope)	504	150	126	228
CS1061	Type does not contain definition	495	102	151	242
CS0246	Type/namespace name not found	469	0	320	149
CS0117	Type does not contain definition (field)	435	93	192	150
CS0103	Name does not exist in context	383	33	272	78
CS1503	Cannot convert argument type	283	63	106	114
CS0176	Static member via instance reference	93	0	87	6
CS0120	Non-static field via type reference	60	54	0	6

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AI in Computational Creativity and Game Design

-Compile rate alone as a misleading evaluation signal.

-A three-factor interpretation, anchored in the error taxonomy.

1- **Domain Completeness:** what information the intermediate representation carries

2- **API-mapping adequacy:** whether the prompt translates intermediate representation entity names into valid Unity API calls.

3- **LLM execution fidelity:** whether the model uses the information correctly

Any LLM code-generation task targeting a domain-specific framework faces these three factors.

Research Themes

- Deep Learning and Audio
- AI in Computational Creativity and Game Design
- **Societal Impact of AI in Culture, Arts, and Music**
- Artworks

Societal Impact of AI in Culture, Arts, and Music

- **A Shift in Artistic Practices through Artificial Intelligence**
- Caring Trouble and Musical AI
- Bringing the Body Back to AI Voice and Speech Technologies

A Shift in Artistic Practices through Artificial Intelligence

Labor in Artistic Data

Infinite Content on the Boundless Internet

Datasets through Internet Scraping

Easily accessible content generators

James Gurney mentions

> "I think it is only fair to people looking at this work that they should know what the prompt was and also what software was used. I think the artists should be allowed to opt-in or opt-out of having their work that they worked so hard on by hand be used as a dataset for creating this other artwork."

Shifts in Artistic Practices

Expansion of Actors involved

New Markets and Economy of Arts and Music

Shifts if Stakeholders and Gatekeepers

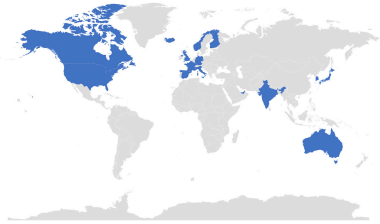
Sustainability, Ethics, Accessibility, and Inclusion

Consent, tracebility, regulations, and accountability

Legal discussions such as copyright

Unequal power in big tech

Countries with AI guidelines



K. Tatar et al., “A Shift in Artistic Practices through Artificial Intelligence,” Leonardo, pp. 293–297, Apr. 2024, doi: 10.1162/leon_a_02523.

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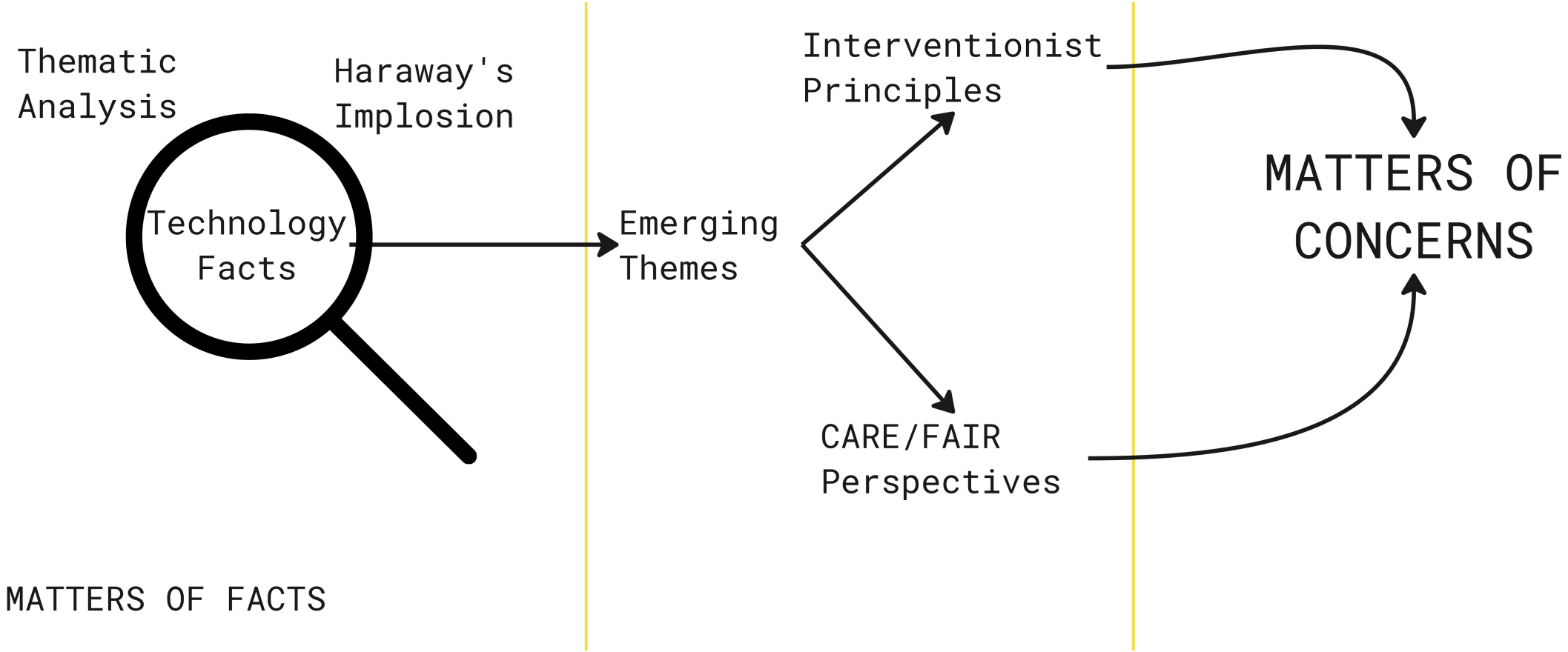
Societal Impact of AI in Culture, Arts, and Music

- A Shift in Artistic Practices through Artificial Intelligence
- **Caring Trouble and Musical AI**
- Bringing the Body Back to AI Voice and Speech Technologies

Caring Trouble and Musical AI



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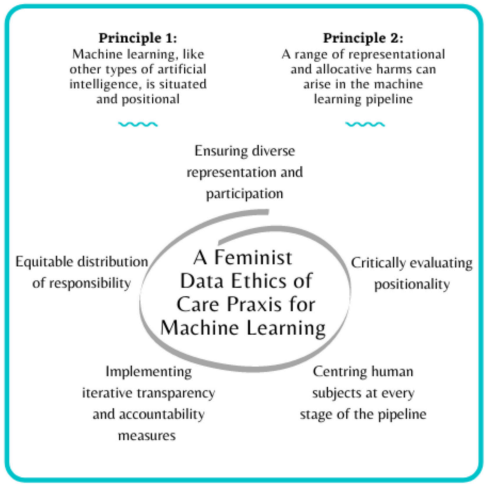
Caring Trouble and Musical AI

Implosion Dimensions

- Labour
- Professional/ Epistemological
- Material
- Technological
- Context and Situatedness
- Political
- Economic
- Textual
- Bodily/Organic
- Historical
- Particle
- Mythological
- Symbolic

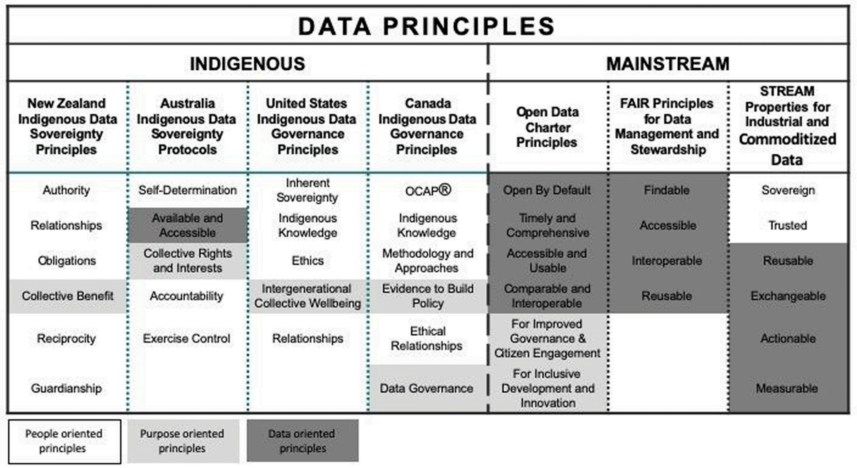
Dumit, J. (2014). Writing the Implosion: Teaching the World One Thing at a Time. Cultural Anthropology, 29(2), 344–362. https://doi.org/10.14506/ca29.2.09

Interventionist Principles



Gray, J., & Witt, A. (2021). A feminist data ethics of care for machine learning: The what, why, who and how. First Monday. https://doi.org/10.5210/fm.v26i12.11833

CARE/FAIR Perspectives



Carroll, S., Garba, I., Figueroa-Rodríguez, O., Holbrook, J., Lovett, R., Materechera, S., … Hudson, M. (2020). The CARE Principles for Indigenous Data Governance. Data Science Journal, (19), 1–12. https://doi.org/https://doi.org/10.5334/dsj-2020-043

Haraway, D. (2013). A cyborg manifesto: Science, technology, and socialist-feminism in the late twentieth century. In The transgender studies reader (pp. 103–118). Routledge.

Societal Impact of AI in Culture, Arts, and Music

-A Shift in Artistic Practices through Artificial Intelligence

-Caring Trouble and Musical AI

-**Bringing the Body Back to AI Voice and Speech Technologies**

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Bringing the Body Back to AI Voice and Speech Technologies

Susan Bennett and Jon Briggs provided voice recordings for GM Voices, which were later licensed to ScanSoft.

Their voice datasets were then later allegedly used to build the voice of the American Siri (Bennett) and British Siri (Briggs) through speech concatenation.

Apple has never confirmed, nor denied whether they utilised Bennett's concatenated speech data, nor Briggs'. In the case of Bennett, audio forensics expert Ed Primeau studied recordings of Siri and blind recordings of Bennett's voice and presented his the conclusion of his analysis that "They are identical – a 100 % match."

Cotton, Kelsey, de Vries, Katja & Tatar, Kıvanç. (2024). Singing for the Missing: Bringing the Body Back to AI Voice and Speech Technologies. In Proceedings of the 9th International Conference on Movement and Computing. Utrecht, Netherlands, May 2024. https://dl.acm.org/doi/10.1145/3658852.3659065

Bringing the Body Back to AI Voice and Speech Technologies

-Integrating voice rights into personality rights frameworks

-The data frameworks for voice should be designed to have direct connection to their source body

-Similarity should be revisited, where nations take their own stances through democratic processes

Research Themes

- Deep Learning and Audio
- AI in Computational Creativity and Game Design
- Societal Impact of AI in Culture, Arts, and Music
- **Artworks**

Artworks

-Expert Procrastinator's Tool: Artificial Intelligence (2023)

-Conceptual Pillars of Artistic Creativity

-Digital Ripples (2020)

-Experiments with VQGAN Text-to-Image Synthesis

-Exposing the Bias in Artificial Intelligence: The Cyber Future

-Exposing the Bias in Artificial Intelligence: The Machine Lexicon

Expert Procrastinator's Tool: Artificial Intelligence (2023)

https://www.youtube.com/watch?v=xdf1uKzGYfs

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Digital Ripples (2020)

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Experiments with VQGAN Text-to-Image Synthesis (2021)

Exposing the Bias in Artificial Intelligence: The Cyber Future

https://www.youtube.com/watch?v=79gJtoeOhHE&source_ve_path=MTc4NDI0

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Exposing the Bias in Artificial Intelligence: The Machine Lexicon

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Exposing the Bias in Artificial Intelligence: The Digital Assemblage

https://youtu.be/cODIggxDDv4

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Research Themes

-Deep Learning and Audio

-Multimodal Deep Learning for Movement and Audio

-AI in Computational Creativity and Game Design

-Societal Impact of AI in Culture, Arts, and Music

-Artworks

Key Takeaways

-How do we make music with vast sound search spaces?

-Most musical AI research are often ad-hoc efforts.

-Music and arts research need its own lineage.

-Technology is not neutral, it is situated in social and cultural contexts.

-It is time for artists and musicians to take on and lead the musical AI technology development.

Future is Bright!

-Mathematics of Latent Audio Spaces for interactivity

-Establishing methods and frameworks for joint efforts in musical AI research

-Moving machines as musical agents

-Creativity beyond the material

-New mixed research methods for analyzing demographic representations in generative AI

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Thank you!

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